Troubleshooting Edge Lifting and Minor Warping in Entry-Level PLA FDM Printing

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Contents

Chapter 1. Before You Begin: Scope and Applicability

This guide is for beginner-to-intermediate desktop FDM users who can clean the build plate, run a first-layer test, and adjust basic slicer settings such as build plate temperature or brim settings.

Use the table below to confirm whether this troubleshooting path matches your situation.

Scope decision table

Condition	**Use this guide?**	**What to do next**
Entry-level desktop FDM printer	Yes	Continue
Flat heated build plate	Yes	Continue
PLA filament	Yes	Continue
Edge or corner lifting during printing	Yes	Continue
Full detachment in the first layers	Partly	Start with first-layer checks
One model fails repeatedly	Partly	Check geometry after basic checks
No extrusion or failed startup	No	Check printer startup first
Temperature error, failed heating, or unstable hardware temperature	No	Check printer-specific hardware guidance
Damaged or loose build plate	No	Fix the build plate first
Non-PLA material	No	Use material-specific guidance
Resin printer	No	Use resin-printing guidance
Manufacturer guidance conflicts with this guide	No	Follow manufacturer guidance

If more than one condition applies, address any hardware, temperature-control, or build plate issue before changing slicer settings. Follow manufacturer documentation when it provides product-specific instructions.

Chapter 2. Quick checklist for edge lifting troubleshooting

Start with this checklist before changing advanced slicer settings.

Recommended troubleshooting order

Step	Check	Why it matters	Typical action
1	Build plate condition	Surface contamination is a common first-layer issue.	Clean the build plate.
2	First-layer contact	Poor first-layer contact often causes early lifting.	Recheck bed leveling or first-layer calibration.
3	Build plate temperature	Weak adhesion may appear after printing starts.	Confirm the PLA profile and build plate temperature.
4	Edge adhesion	Narrow contact areas lift easily.	Add a brim if needed.
5	Surrounding airflow	Uneven cooling can make one side lift more easily.	Reduce drafts and unstable cooling.
6	Part geometry	Some shapes create more shrinkage stress.	Review print orientation first.
7	Secondary slicer settings	These are not first-line fixes.	Test bottom surface or infill pattern changes only after the basic checks.

Symptom-to-check guide

If you see this	Check this first	Go to
Adhesion is inconsistent across the build plate	Build plate condition	Build plate surface *(on page 6)*
Small prints lift even when settings appear normal	Build plate condition	Build plate surface *(on page 6)*
One corner lifts more than the others	Airflow and uneven cooling	Airflow and room conditions *(on page 10)*
One side of the model sticks better than the other	First-layer calibration and bed leveling	First-layer height *(on page 7)*

If you see this	Check this first	Go to
Large flat models fail after a normal start	Build plate temperature, edge adhesion, and part geometry	Build plate temperature *(on page 8)*, Edge adhesion *(on page 9)*, then Part geometry *(on page 11)*
Narrow-base parts detach more easily	Edge adhesion	Edge adhesion *(on page 9)*
The same model sometimes succeeds and sometimes fails	Room conditions and airflow	Airflow and room conditions *(on page 10)*
Large flat models still fail after all basic checks	Secondary slicer settings such as bottom surface pattern or infill pattern	Secondary slicer adjustments *(on page 12)*

If your situation matches more than one row, start from the earliest step in the recommended troubleshooting order.

Chapter 3. Fix edge lifting and minor warping in PLA prints

If a PLA print starts normally but the lower edges or corners lift from the build plate during printing, follow these steps to diagnose and resolve the issue.

Note: Change only one setting or environmental factor at a time. After each change, test under the same conditions before changing another factor.

CAUTION:

The nozzle and heated build plate can remain hot before, during, and after printing. Let the printer cool down, or follow the manufacturer's safety guidance before cleaning the build plate or making manual adjustments.

Build plate surface is dirty or poorly prepared

Cause

Dust, oil, fingerprints, or leftover residue on the build plate can reduce first-layer adhesion.

Common symptoms

- Small prints lift even when settings appear normal.
- Adhesion is inconsistent across the build plate.
- A print starts normally but loses contact later.

Recommended action

- Clean the build plate according to the manufacturer's care guidance.
- Do not touch the print area after cleaning.
- Confirm that the selected build surface profile, or the printer or slicer setting for the installed plate type, matches the installed build surface.

Check the result

- Restart the same print or run the same first-layer test under the same conditions.

- If the first layer stays attached across the print area, stop here and keep the build plate preparation method.

- If adhesion is still inconsistent, continue to first-layer height or bed leveling checks.

Notes

- Cleaning methods vary by build surface.

First-layer height or bed leveling is incorrect

Cause

If the nozzle is too close to the build plate or too far from it, the first layer may not form stable contact.

Common symptoms

- Extruded lines look too round and loose.
- Extruded lines look too thin and overly squashed.
- One side of the model sticks better than the other.
- Edge lifting starts from one area of the build plate.

Recommended action

- Recheck bed leveling or first-layer calibration.
- Use a simple first-layer test print if one is available.
- Adjust the nozzle-to-build-plate distance in small increments. On printers that provide a Z-offset or other micron-based height adjustment, about 0.02 mm can be used as a cautious starting point.
- After each adjustment, repeat the same first-layer test before starting another full print.

Check the result

- Repeat the same first-layer test after each adjustment.

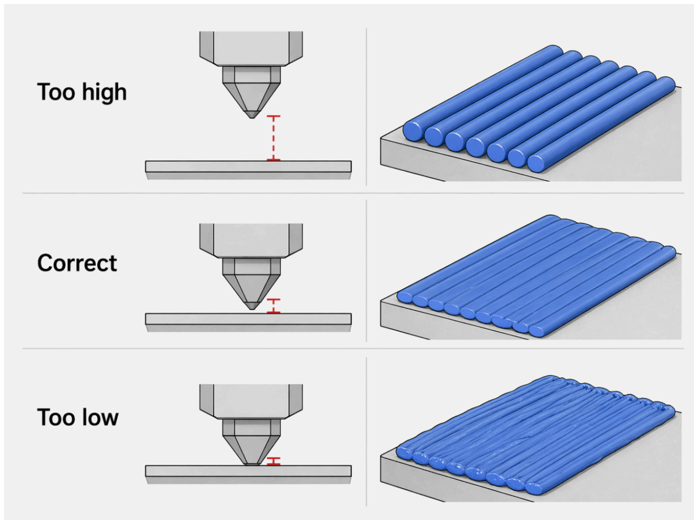
- If the first-layer lines show stable contact across the build plate, stop here and keep the current calibration.

- If one side still sticks better than the other, recheck bed leveling before changing slicer settings.

Notes

- A first-layer calibration print or first-layer test print is usually a more reliable way to confirm contact.

Figure 1. First-layer line shape changes when the nozzle is too high, correctly set, or too low.



Build plate temperature is too low or unstable

Cause

If the build plate is too cool, the first layer may not bond strongly enough to resist shrinkage stress at the edges. If the build plate temperature is unstable, adhesion may also become less consistent during printing.

Common symptoms

- The print starts normally but corners lift later.
- Larger flat models fail more easily than smaller ones.
- Prints fail more often in cooler conditions.

Recommended action

- For PLA on a heated bed, 50–60°C is a common reference range. If the manufacturer recommends a different value, follow that value.

- If corners lift after the print starts and the current value is near the lower end of the recommended range, test a slightly higher build plate temperature under the same conditions.

Check the result

- Restart the same print after changing only the build plate temperature.
- If the lower edges stay attached through the early layers, stop here and keep the tested temperature.
- If corners still lift, return to the previous temperature before testing another factor.

Notes

- A higher build plate temperature is not a universal fix.
- This factor often interacts with first-layer contact, part size, and room conditions.

Edge adhesion is insufficient

Cause

The base of the model may not have enough contact area to resist upward pulling force.

Common symptoms

- Corners lift first even when the center still holds.
- Narrow-base parts detach more easily.
- Tall parts with a small footprint become unstable.

Recommended action

- Add a brim to increase contact area around the model.
- Start with the slicer's default brim setting or a narrow brim, and increase the width one step at a time only if edge lifting continues.
- Use a brim before moving to more advanced slicer changes.

Check the result

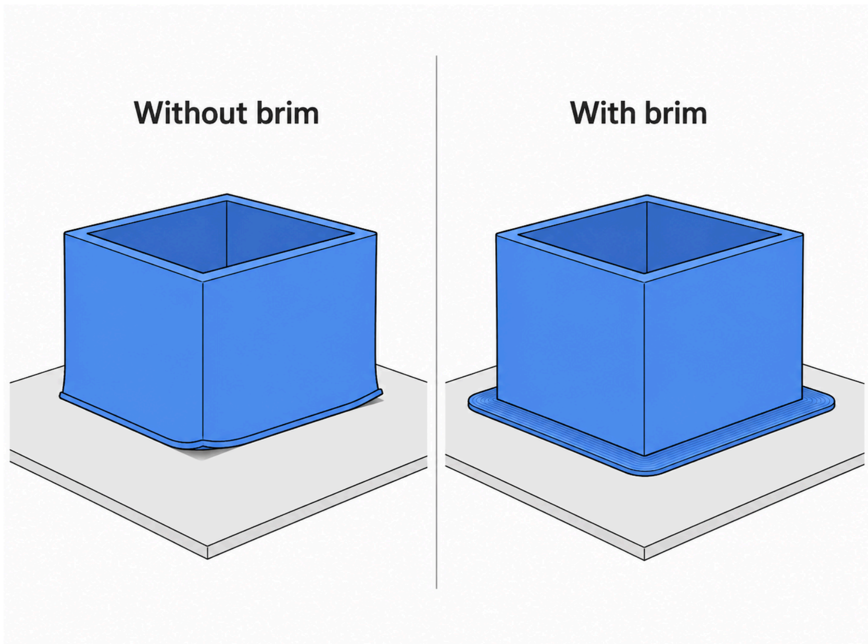
- Restart the same print with the brim enabled.
- If the corners and narrow contact areas stay attached, stop here and keep the brim setting.
- If lifting continues, increase the brim width one step at a time.

- If a wider brim does not improve the result, check airflow, room conditions, or part geometry instead.

Notes

- A brim may leave extra material around the model base that must be removed after printing.

Figure 2. A brim increases contact area around the model base and helps resist edge lifting.



Airflow or room conditions cause uneven cooling

Cause

Drafts or unstable room temperature can cool one side of the print faster than the other. Uneven cooling can increase shrinkage stress along one edge or corner.

Common symptoms

- One corner lifts more than the others.
- The same model sometimes succeeds and sometimes fails.
- Problems occur more often near windows, fans, or air vents.

Recommended action

- Reduce direct airflow around the printer.
- Avoid sudden airflow or temperature changes during printing.
- Avoid placing the printer near open windows, fans, air vents, or other strong moving air sources.

Check the result

- Restart the same print after reducing direct airflow around the printer.
- If the same corner or side no longer lifts under similar conditions, stop here and keep the more stable airflow setup.
- If the issue appears only at certain times or locations, review nearby airflow sources again.

Notes

- For PLA printing, a full enclosure is not required in many normal cases.
- Small changes in airflow can still affect build plate adhesion, especially on larger flat parts.
- More stable room conditions may help reduce repeated edge lifting.

Part geometry or orientation increases shrinkage stress

Cause

Large flat bases, long straight edges, and sharp corners can concentrate shrinkage stress during cooling.

Common symptoms

- Long rectangular parts lift at both ends.
- Sharp corners lift more easily than rounded ones.
- Lifting begins along long exposed edges on otherwise well-adhered parts.

Recommended action

- Reconsider the print orientation if possible.
- Avoid placing long straight edges directly along the build plate when another orientation is possible.
- Reduce long exposed edge length when the model allows it.

Check the result

- Test the same model in the revised orientation when possible.
- If edge lifting is reduced under the same printing conditions, stop here and use the revised orientation for this model.
- If orientation changes do not reduce lifting, return to the basic adhesion checks or consider model-specific design changes.

Notes

- Some shapes create more stress even when first-layer calibration and build plate adhesion are acceptable.

Use secondary slicer adjustments only after the basic checks

Use the following slicer-level adjustments only after checking build plate condition, first-layer contact, build plate temperature, edge adhesion, airflow, and part geometry.

These adjustments do not replace model-specific design changes when the model shape itself causes repeated lifting.

Bottom surface pattern may affect stress distribution

Cause

On some large flat models, the bottom surface pattern may influence how cooling stress becomes visible across the base.

Common symptoms

- Large flat models continue to show edge lifting after all basic checks.
- Edge lifting appears in a similar location each time when the same slicer profile is used.

Recommended action

- Test one alternative bottom surface pattern, such as Concentric or Zigzag, on the same model.

Check the result

- Test only one alternative bottom surface pattern on the same model.
- If edge lifting is reduced under the same printing conditions, stop here and keep the tested bottom surface pattern for this model.
- If the result does not improve, return to the previous pattern before testing another slicer setting.

Notes

- On some flat parts, line-based paths may create longer continuous stress paths across the base.
- No bottom surface pattern works best for every model.

Infill pattern may affect internal stress behavior

Cause

The infill pattern may affect how cooling stress builds inside the part after the base is already attached.

Common symptoms

- Minor warping remains after first-layer correction and brim use.
- The base adheres well at first, but warping develops as the print grows taller.

Recommended action

- Test an alternative infill pattern on the same model.

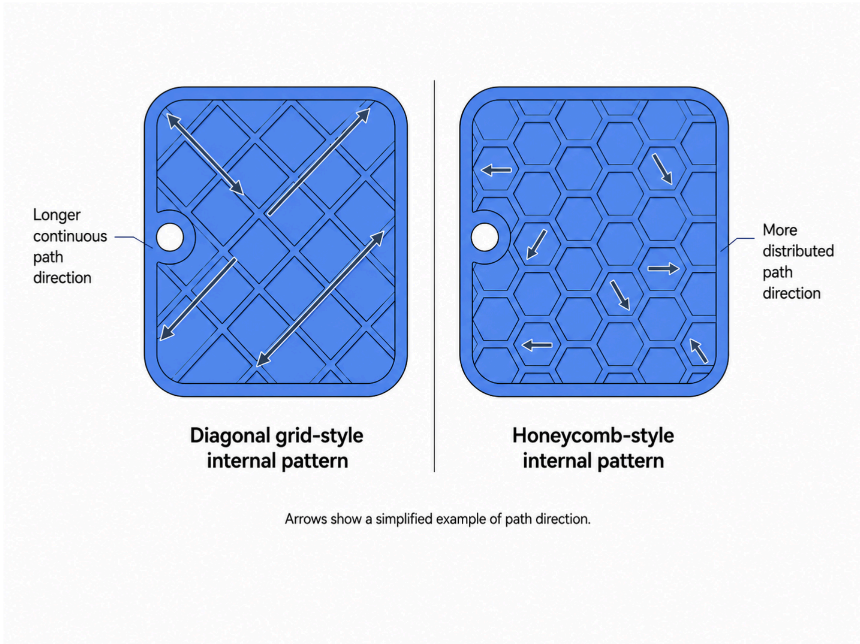
Check the result

- Test only one alternative infill pattern on the same model.
- If minor warping is reduced after the base has already adhered, stop here and keep the tested infill pattern for this model.
- If the result does not improve, return to the previous infill pattern before testing another slicer setting.

Notes

- Some infill patterns create longer crossing paths or a different material distribution inside the part.
- Results vary by model.

Figure 3. Internal patterns can change how stress paths are distributed inside a part.



If the problem persists

If edge lifting continues after the basic checks, stop changing slicer settings and narrow down the next likely cause.

Escalation table

If this happens	Next area	Action
First-layer lines still look uneven	First-layer calibration	Follow the printer-specific calibration workflow
No extrusion or failed startup appears	Printer startup	Check printer startup guidance
The build plate cannot hold the target temperature	Temperature control	Check hardware guidance
The build plate is damaged, loose, or heavily worn	Build plate condition	Fix the build plate first
The same PLA spool fails on simple models	Filament condition	Try a known good PLA spool
Only one model fails repeatedly	Model geometry	Review orientation and shape

If this happens	**Next area**	**Action**
The same setup fails inconsistently	Airflow or surface condition	Check airflow and clean the build plate again
Large flat models still fail after the basic checks	Secondary slicer settings	Test one secondary slicer adjustment at a time
The issue starts after a slicer profile change	Slicer profile	Compare with a known good profile

Stop here if the issue appears to be hardware-related. Follow printer-specific or slicer-specific documentation before making more changes.

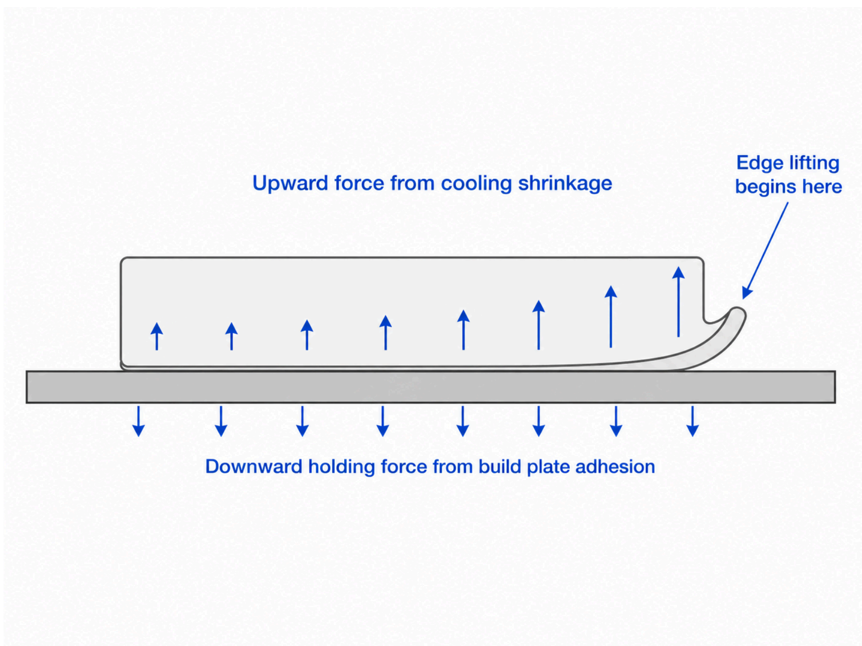
Chapter 4. Why PLA prints develop edge lifting and minor warping

Why the problem happens

PLA cools and shrinks slightly as each layer is deposited. That shrinkage creates an upward pulling force on the lower edges of the part.

The build plate must hold the part down while this happens. When the shrinkage force becomes stronger than build plate adhesion, the edges begin to lift.

Figure 4. Edge lifting begins when cooling shrinkage overcomes build plate adhesion at the edge.



Why corners usually lift first

Corners often lift first because they have less surrounding material and more exposure to open-air cooling. Once a corner lifts, the nearby edge often follows.

Why large flat parts warp more easily

Large flat parts create more total shrinkage stress as PLA cools.

Long straight edges can also accumulate stress in one direction. This can make the lower perimeter more likely to pull away from the build plate.

For that reason, parts that look simple because they have a large flat base may actually be harder to keep fully attached during printing.

Why the first layers matter most

The first layers determine how well the rest of the part stays anchored to the build plate.

If first-layer contact is weak, later shrinkage forces become more visible. As the print grows taller, the lower edges may be pulled upward more easily.

For that reason, edge lifting and minor warping often begin as an early first-layer problem, even when the print continues for some time afterward.

Chapter 5. Key terms for PLA edge lifting troubleshooting

Term	Definition
First layer	The first printed layer placed directly onto the build plate. Its quality affects how well the rest of the part stays attached during printing.
First-layer calibration	The process of checking and adjusting first-layer contact so that the printed lines bond correctly to the build plate.
Build plate adhesion	The ability of a printed part to stay attached to the build plate while it is being printed.
Brim	A thin ring of extra printed material added around the base of a part to increase contact area.
Z-offset	A printer setting that adjusts the nozzle height relative to the build plate. It is often used to fine-tune first-layer contact.
Build surface profile	The slicer or printer setting that matches the installed build surface or plate type.
Bottom surface pattern	The path strategy used for the solid bottom layers where the print meets the build plate.
Infill pattern	The path pattern used to print the internal support structure inside a part.

Note: Bottom surface pattern affects the solid lower surface. Infill pattern affects the internal structure.